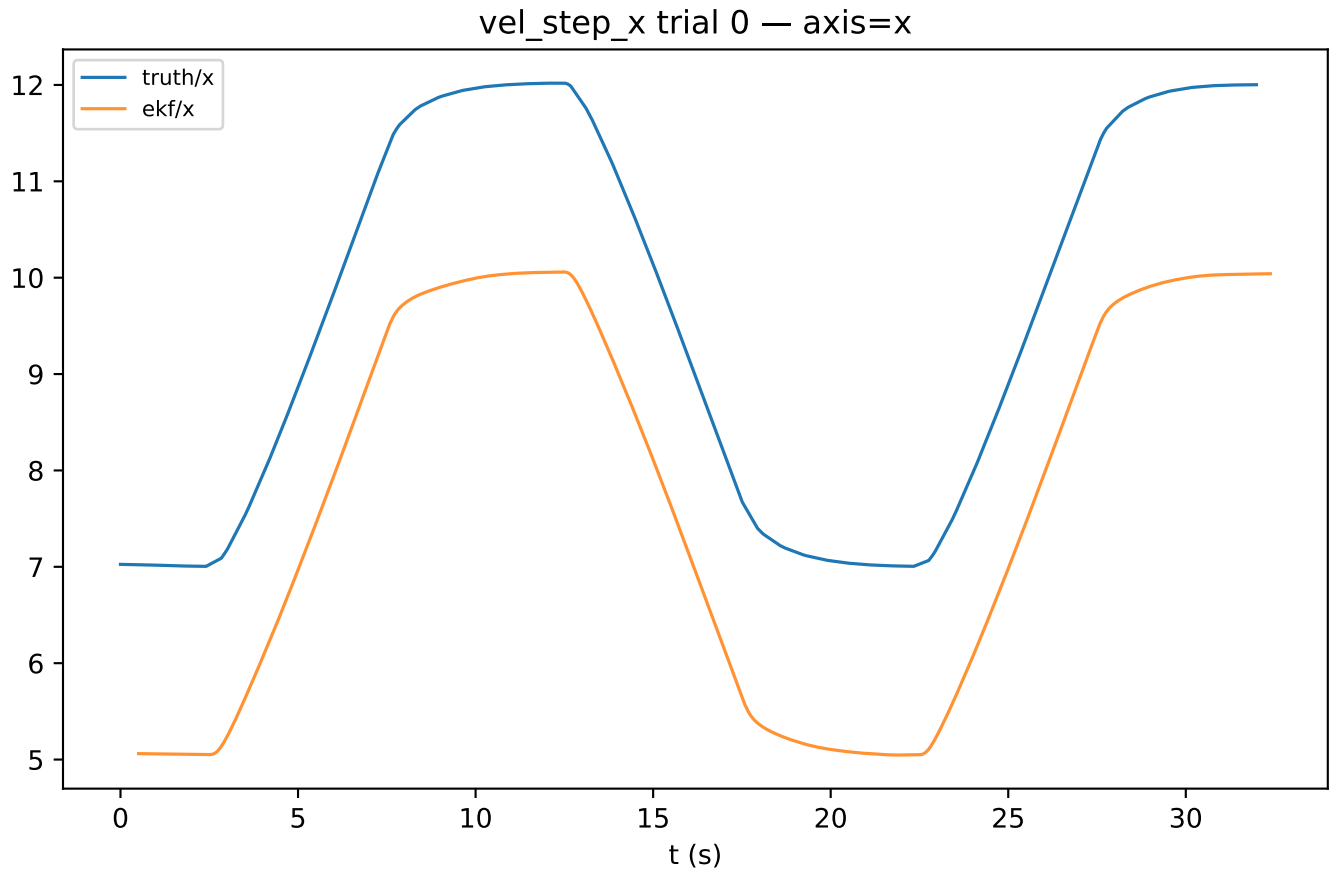


position_xyz status=fit delay_mean=0.0000s xcorr_peak=1.00



channel: position_xyz

source_maneuvers: ['vel_step_x/trial0', 'vel_step_x/trial1', 'vel_step_x/trial2', 'vel_step_z/trial0', 'vel_s

x:

delay_xcorr_mean_s: 0.00000
delay_xcorr_std_s: 0.00000
xcorr_peak_mean: 0.99979
delay_step50_mean_s: 0.05613
delay_chirp_mean_s: None
chirp_flatness_rel_mean: None
residual_rmse_mean: None
n_xcorr_samples: 3
n_step50_samples: 6
n_chirp_samples: 0

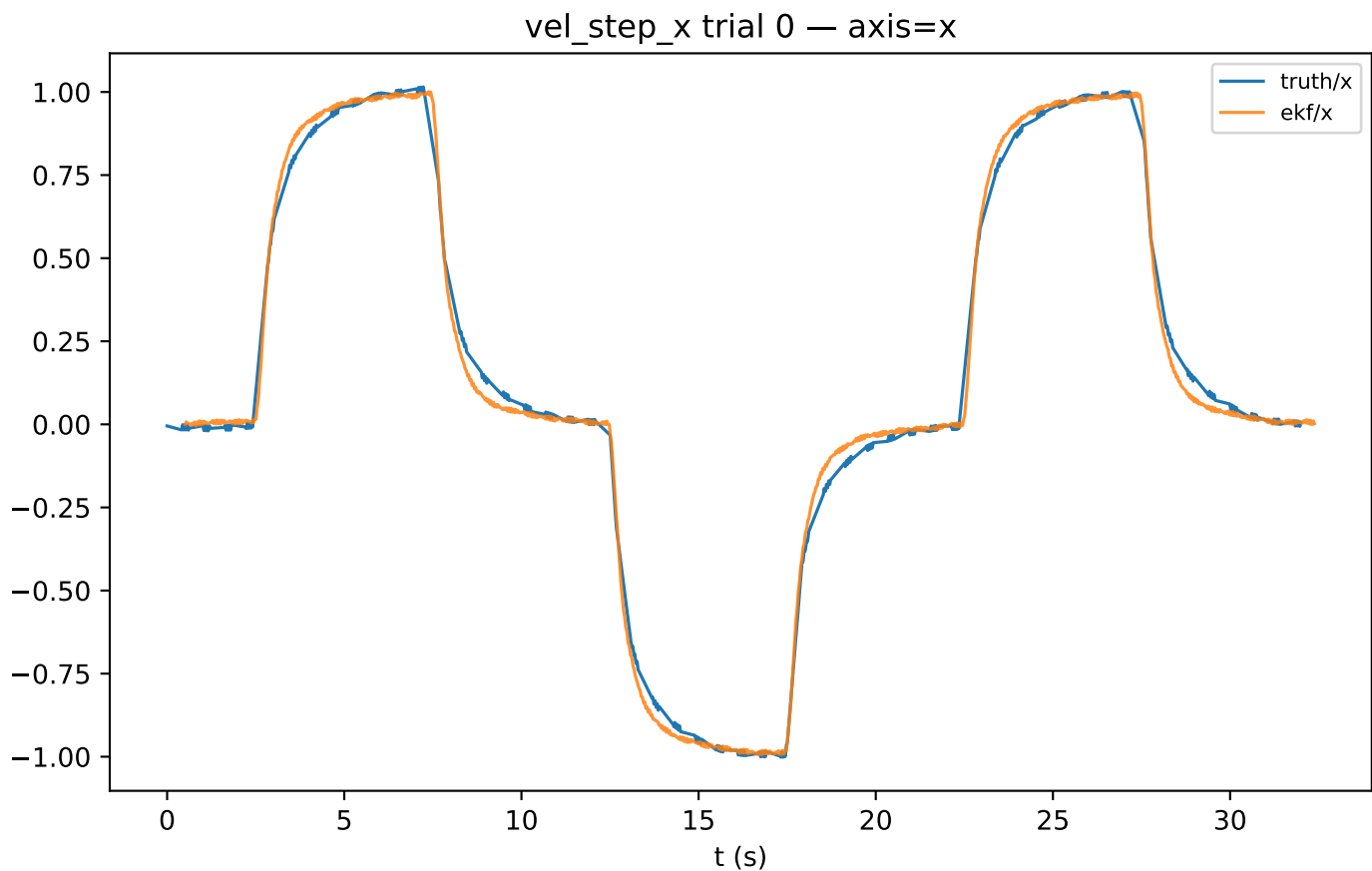
y:

delay_xcorr_mean_s: None
delay_xcorr_std_s: None
xcorr_peak_mean: None
delay_step50_mean_s: 0.40334
delay_chirp_mean_s: None
chirp_flatness_rel_mean: None
residual_rmse_mean: None
n_xcorr_samples: 0
n_step50_samples: 6
n_chirp_samples: 0

z:

delay_xcorr_mean_s: 0.00000
delay_xcorr_std_s: 0.00000
xcorr_peak_mean: 0.99969
delay_step50_mean_s: 0.22000
delay_chirp_mean_s: None
chirp_flatness_rel_mean: None
residual_rmse_mean: None
n_xcorr_samples: 3
n_step50_samples: 6
n_chirp_samples: 0

velocity_world_xyz status=fit delay_mean=-0.0000s xcorr_peak=1.00



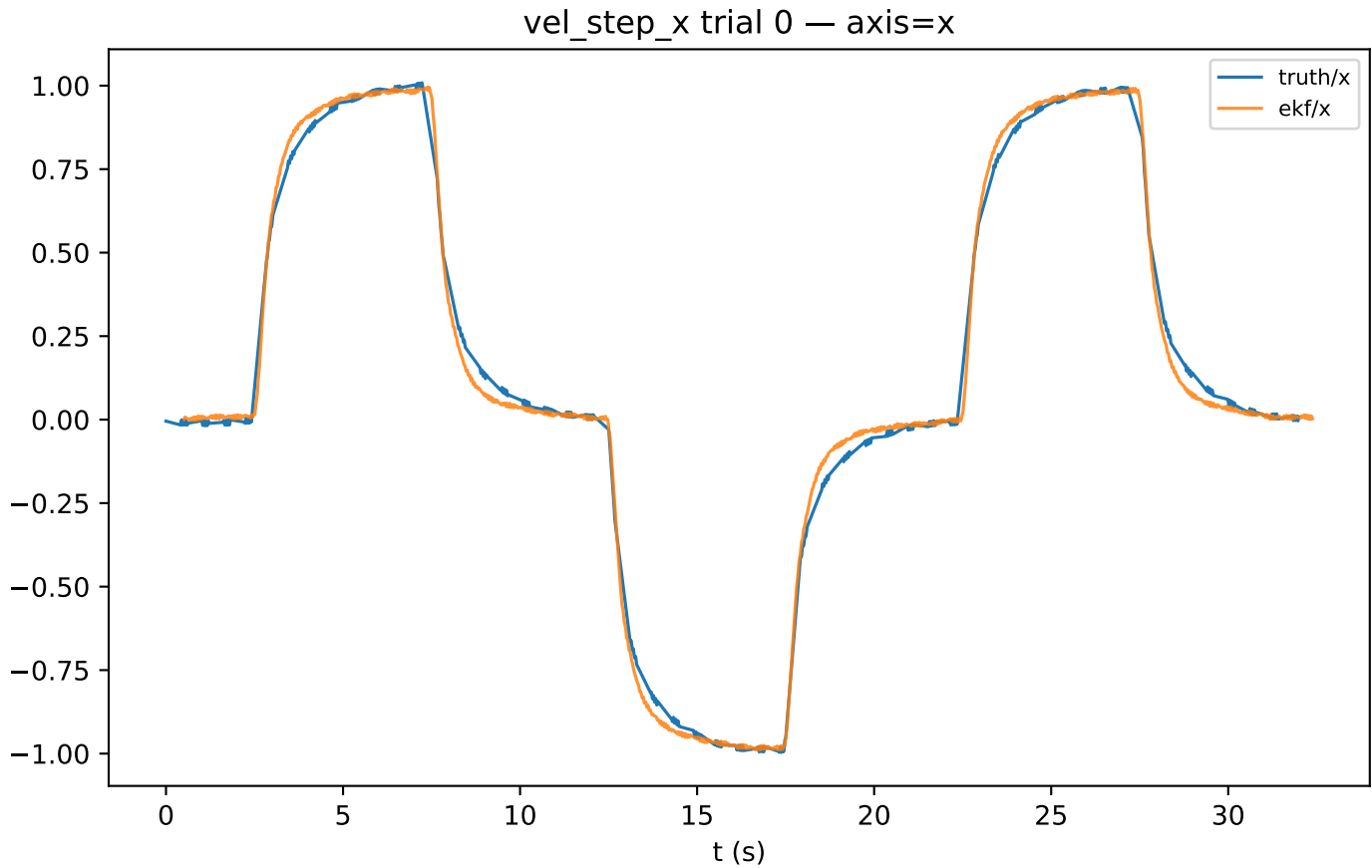
channel: velocity_world_xyz
source_maneuvers: ['vel_step_x/trial0', 'vel_step_x/trial1', 'vel_step_x/trial2', 'vel_step_z/trial0', 'vel_s

x:
delay_xcorr_mean_s: -0.01167
delay_xcorr_std_s: 0.00624
xcorr_peak_mean: 0.99827
delay_step50_mean_s: -0.19425
delay_chirp_mean_s: None
chirp_flatness_rel_mean: None
residual_rmse_mean: None
n_xcorr_samples: 3
n_step50_samples: 6
n_chirp_samples: 0

y:
delay_xcorr_mean_s: None
delay_xcorr_std_s: None
xcorr_peak_mean: None
delay_step50_mean_s: -0.80732
delay_chirp_mean_s: None
chirp_flatness_rel_mean: None
residual_rmse_mean: None
n_xcorr_samples: 0
n_step50_samples: 6
n_chirp_samples: 0

z:
delay_xcorr_mean_s: 0.01167
delay_xcorr_std_s: 0.00236
xcorr_peak_mean: 0.99397
delay_step50_mean_s: -0.62983
delay_chirp_mean_s: None
chirp_flatness_rel_mean: None
residual_rmse_mean: 0.03733
n_xcorr_samples: 3
n_step50_samples: 6
n_chirp_samples: 0

velocity_body_xyz status=fit delay_mean=-0.0000s xcorr_peak=1.00



channel: velocity_body_xyz

source_maneuvers: ['vel_step_x/trial0', 'vel_step_x/trial1', 'vel_step_x/trial2', 'vel_step_z/trial0', 'vel_s

x:

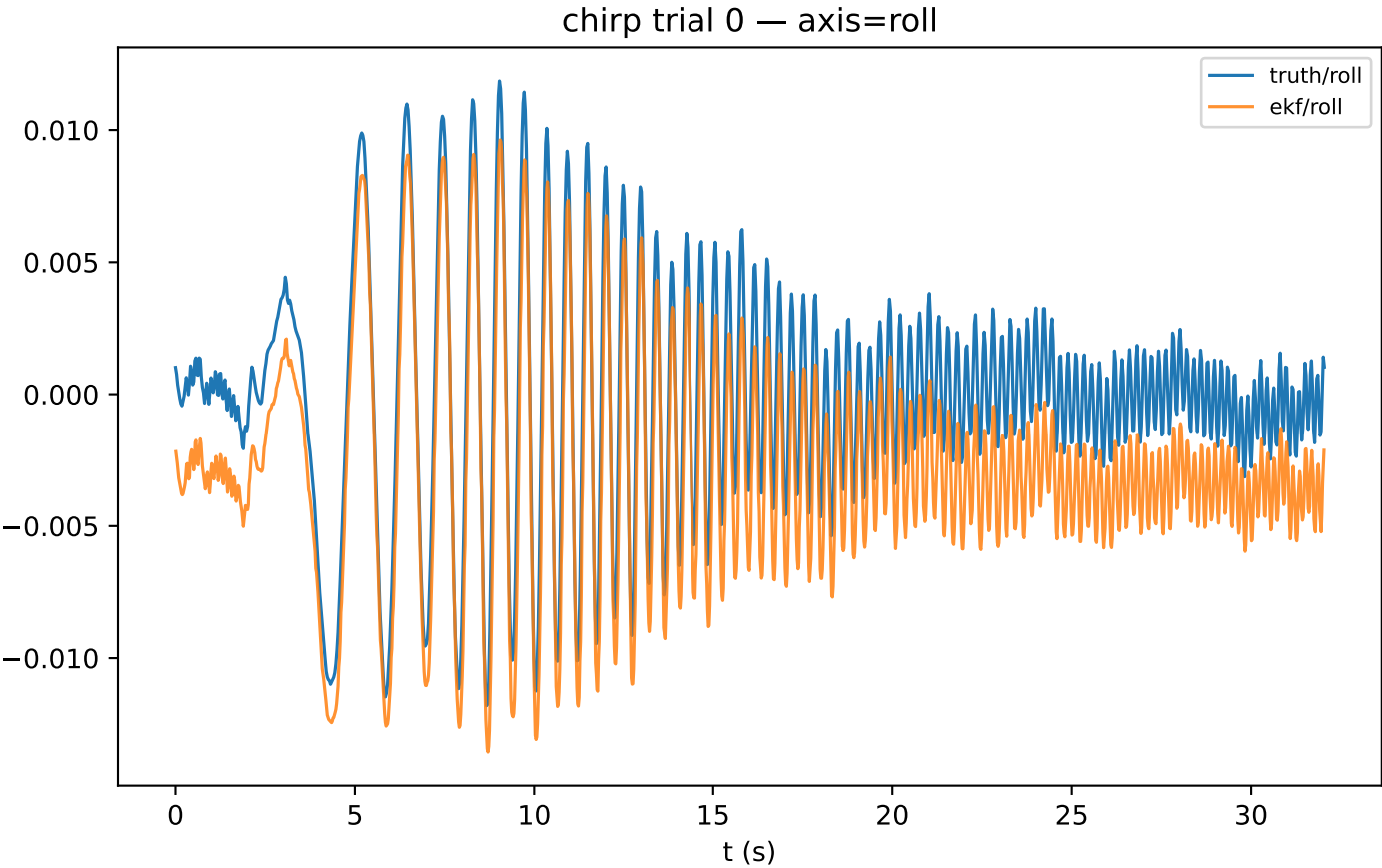
delay_xcorr_mean_s: -0.01167
delay_xcorr_std_s: 0.00624
xcorr_peak_mean: 0.99826
delay_step50_mean_s: -0.24648
delay_chirp_mean_s: None
chirp_flatness_rel_mean: None
residual_rmse_mean: None
n_xcorr_samples: 3
n_step50_samples: 6
n_chirp_samples: 0

y:

delay_xcorr_mean_s: None
delay_xcorr_std_s: None
xcorr_peak_mean: None
delay_step50_mean_s: -0.40222
delay_chirp_mean_s: None
chirp_flatness_rel_mean: None
residual_rmse_mean: None
n_xcorr_samples: 0
n_step50_samples: 6
n_chirp_samples: 0

z:

delay_xcorr_mean_s: 0.01167
delay_xcorr_std_s: 0.00236
xcorr_peak_mean: 0.99397
delay_step50_mean_s: 0.05610
delay_chirp_mean_s: None
chirp_flatness_rel_mean: None
residual_rmse_mean: 0.03733
n_xcorr_samples: 3
n_step50_samples: 6
n_chirp_samples: 0

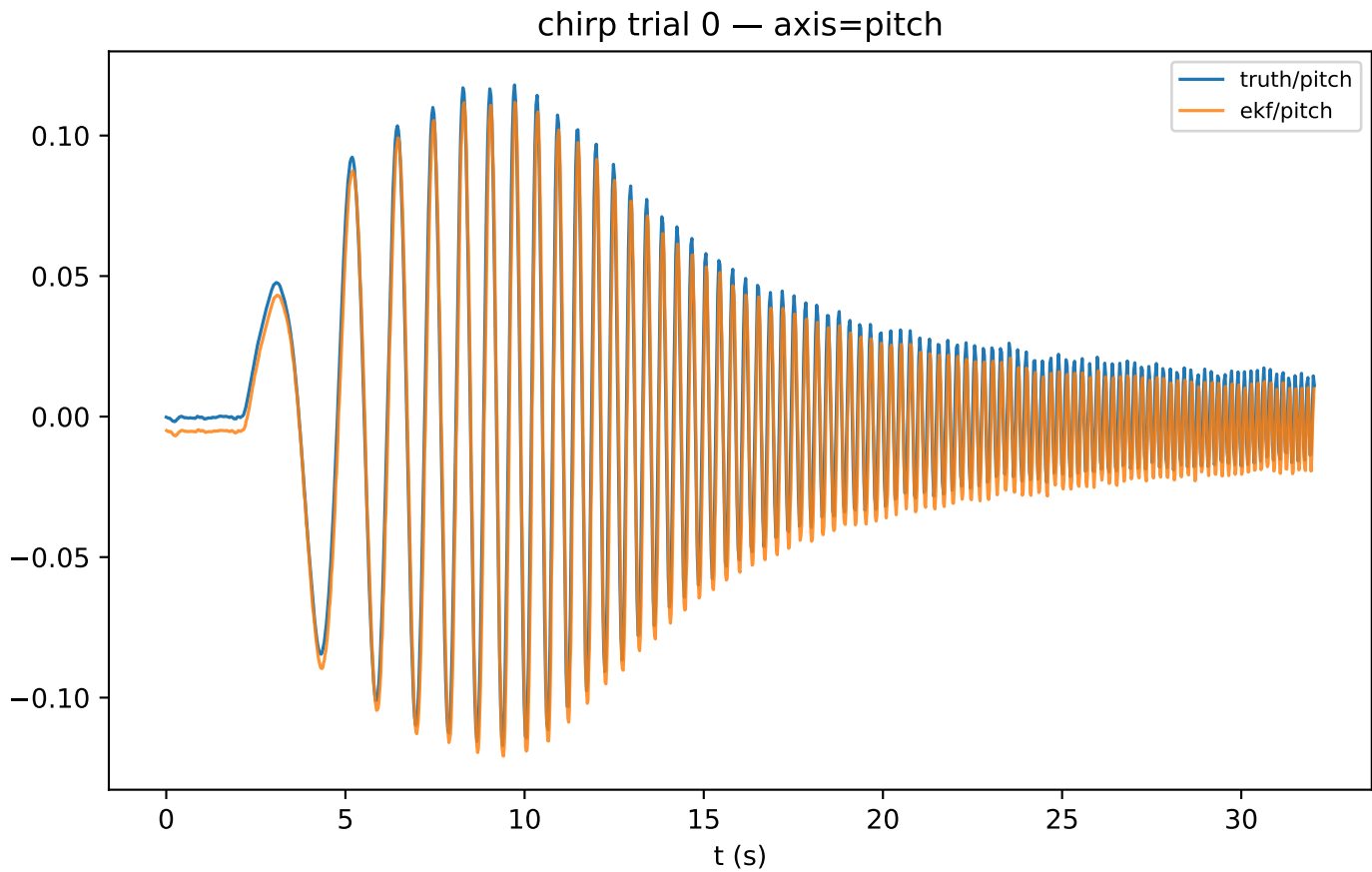


```
channel: attitude_roll
source_maneuvers: ['chirp/trial0', 'chirp/trial1', 'chirp/trial2']

roll:
  delay_xcorr_mean_s: 0.01833
  delay_xcorr_std_s: 0.00236
  xcorr_peak_mean: 0.98743
  delay_step50_mean_s: 0.04552
  delay_chirp_mean_s: 0.32273
  chirp_flatness_rel_mean: 0.44642
  residual_rmse_mean: 0.00376
  n_xcorr_samples: 3
  n_step50_samples: 3
  n_chirp_samples: 3
```

NOTE: chirp group-delay flatness 0.45 > 0.2 – constant-delay model may be insufficient; consider escalating to

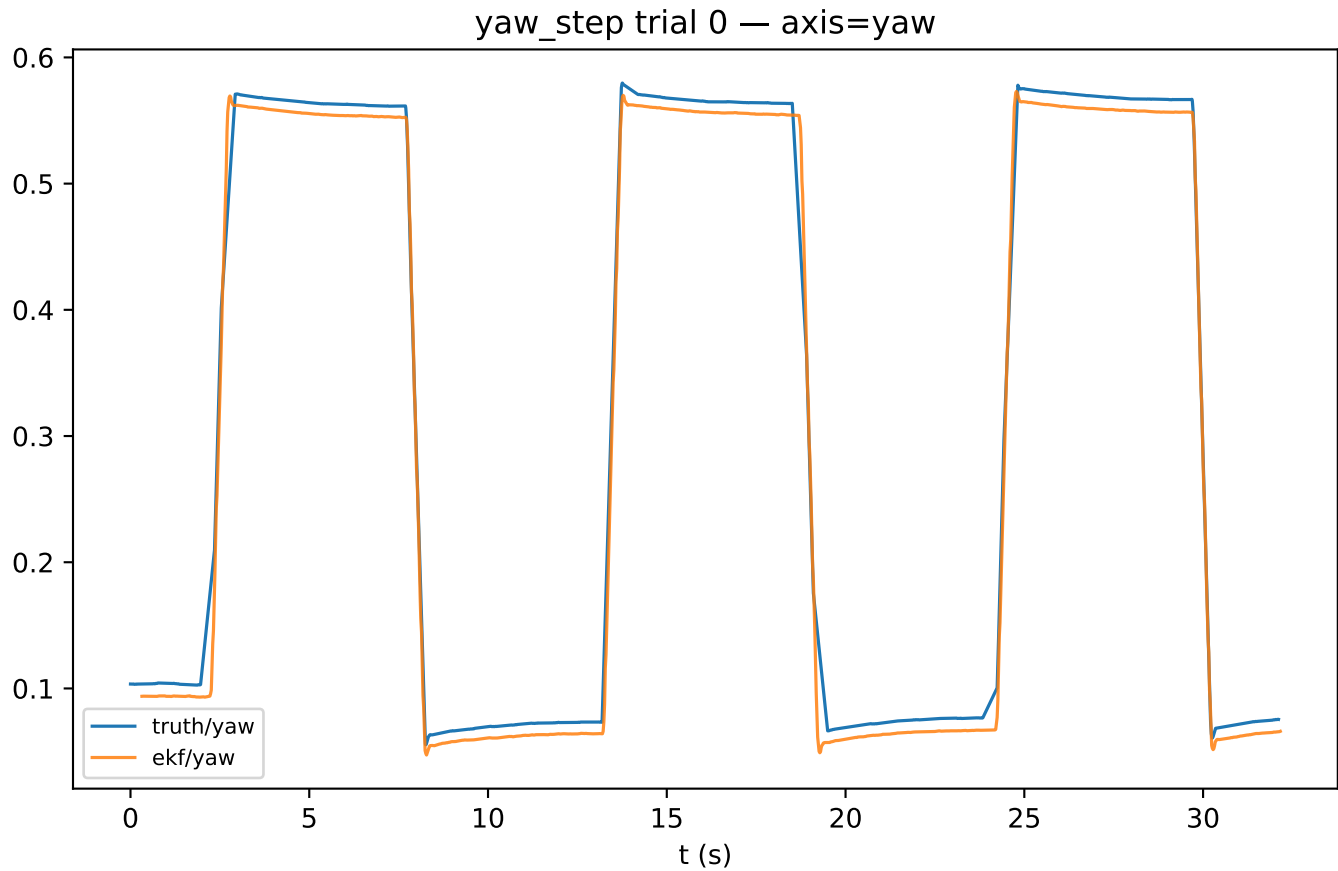
attitude_pitch status=fit delay_mean=0.0175s xcorr_peak=0.98



```
channel: attitude_pitch
source_maneuvers: ['chirp/trial0', 'chirp/trial1', 'chirp/trial2', 'vel_step_x/trial0', 'vel_step_x/trial1',
pitch:
  delay_xcorr_mean_s: 0.01750
  delay_xcorr_std_s: 0.00479
  xcorr_peak_mean: 0.97598
  delay_step50_mean_s: -0.10696
  delay_chirp_mean_s: -0.28392
  chirp_flatness_rel_mean: 0.33138
  residual_rmse_mean: 0.00959
  n_xcorr_samples: 6
  n_step50_samples: 6
  n_chirp_samples: 3
```

NOTE: chirp group-delay flatness 0.33 > 0.2 — constant-delay model may be insufficient; consider escalating t

attitude_yaw status=fit delay_mean=0.0000s xcorr_peak=1.00

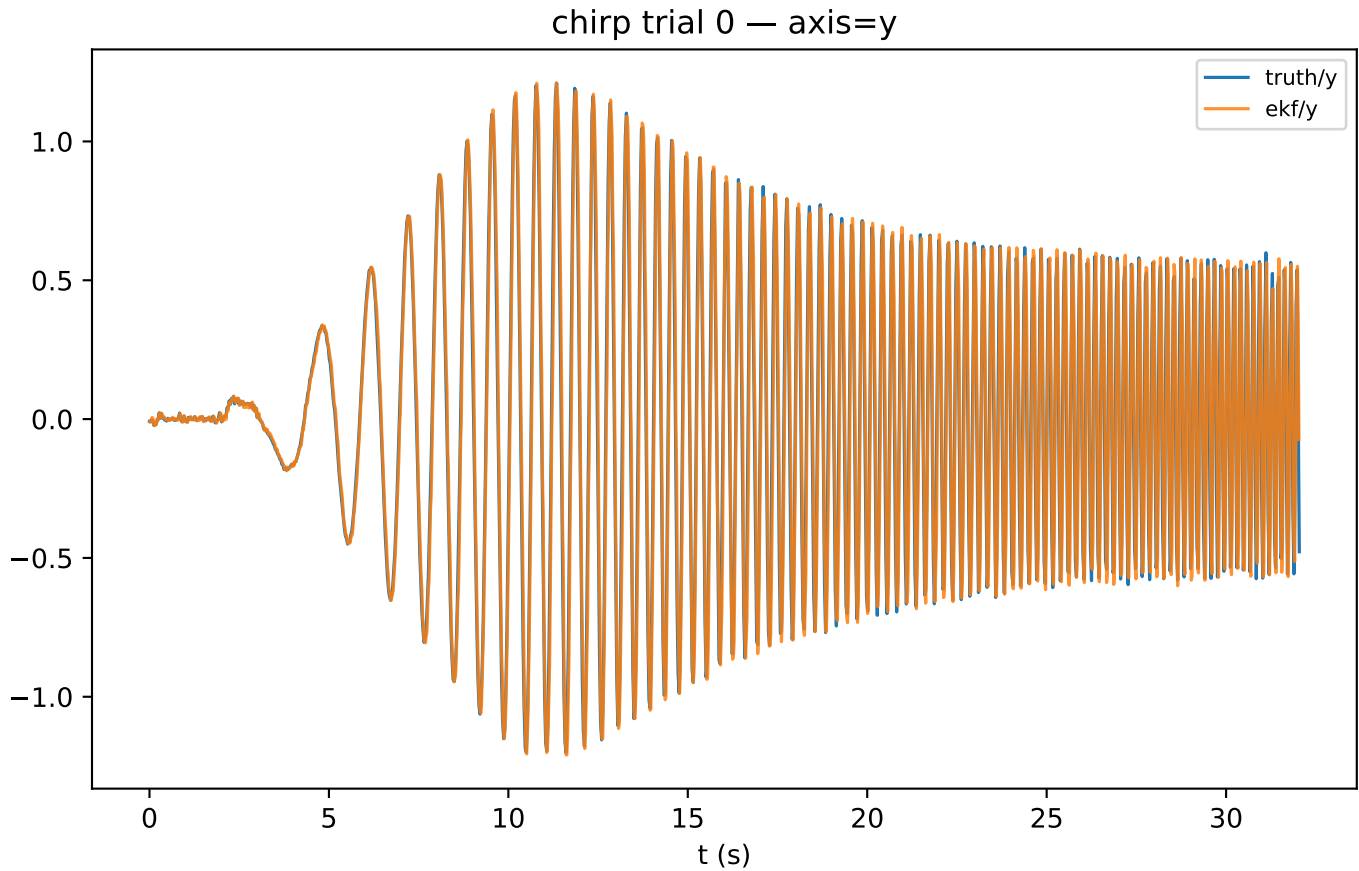


channel: attitude_yaw
source_maneuvers: ['yaw_step/trial0', 'yaw_step/trial1', 'yaw_step/trial2']

yaw:
delay_xcorr_mean_s: 0.00000
delay_xcorr_std_s: 0.00000
xcorr_peak_mean: 0.99826
delay_step50_mean_s: 0.01557
delay_chirp_mean_s: None
chirp_flatness_rel_mean: None
residual_rmse_mean: None
n_xcorr_samples: 3
n_step50_samples: 3
n_chirp_samples: 0

NOTE: delay_mean 0.0 ms (≤ 10 ms) — within measurement resolution; consistent with PX4's lag-compensated EKF output

body_rate_xyz status=fit delay_mean=0.0150s xcorr_peak=1.00



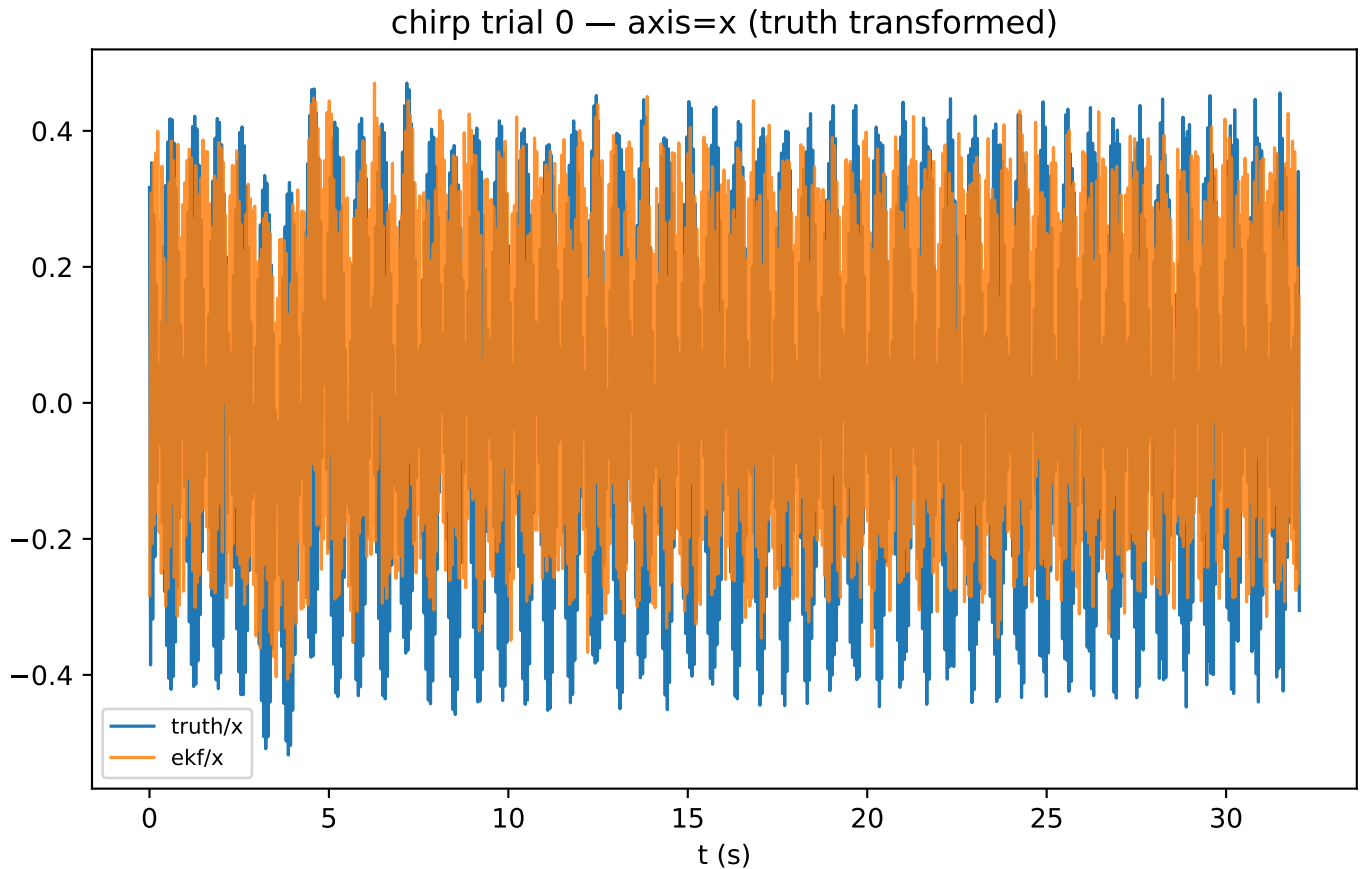
channel: body_rate_xyz
source_maneuvers: ['chirp/trial0', 'chirp/trial1', 'chirp/trial2']

x:
delay_xcorr_mean_s: None
delay_xcorr_std_s: None
xcorr_peak_mean: None
delay_step50_mean_s: -0.16646
delay_chirp_mean_s: 0.01756
chirp_flatness_rel_mean: 2.28697
residual_rmse_mean: None
n_xcorr_samples: 0
n_step50_samples: 6
n_chirp_samples: 3

y:
delay_xcorr_mean_s: 0.01500
delay_xcorr_std_s: 0.00000
xcorr_peak_mean: 0.99734
delay_step50_mean_s: -0.25171
delay_chirp_mean_s: 0.92415
chirp_flatness_rel_mean: 0.68439
residual_rmse_mean: 0.03780
n_xcorr_samples: 3
n_step50_samples: 6
n_chirp_samples: 3

z:
delay_xcorr_mean_s: None
delay_xcorr_std_s: None
xcorr_peak_mean: None
delay_step50_mean_s: -0.08083
delay_chirp_mean_s: 0.93940
chirp_flatness_rel_mean: 0.64600
residual_rmse_mean: None
n_xcorr_samples: 0
n_step50_samples: 6
n_chirp_samples: 3

linear_acceleration_xyz status=fit delay_mean=0.0353s xcorr_peak=0.77



channel: linear_acceleration_xyz
source_maneuvers: ['chirp/trial0', 'chirp/trial1', 'chirp/trial2', 'vel_impulse_recovery/trial0', 'vel_impulse_recovery/trial1', 'vel_impulse_recovery/trial2']

x:
delay_xcorr_mean_s: 0.01833
delay_xcorr_std_s: 0.03064
xcorr_peak_mean: 0.69165
delay_step50_mean_s: None
delay_chirp_mean_s: 1.03801
chirp_flatness_rel_mean: 0.51119
residual_rmse_mean: 0.13728
n_xcorr_samples: 6
n_step50_samples: 0
n_chirp_samples: 3

y:
delay_xcorr_mean_s: 0.05200
delay_xcorr_std_s: 0.02400
xcorr_peak_mean: 0.67107
delay_step50_mean_s: None
delay_chirp_mean_s: 2.86175
chirp_flatness_rel_mean: 0.28557
residual_rmse_mean: 0.07365
n_xcorr_samples: 5
n_step50_samples: 0
n_chirp_samples: 3

z:
delay_xcorr_mean_s: 0.03833
delay_xcorr_std_s: 0.00236
xcorr_peak_mean: 0.92974
delay_step50_mean_s: None
delay_chirp_mean_s: 0.70000
chirp_flatness_rel_mean: 1.11047
residual_rmse_mean: 0.02899
n_xcorr_samples: 6
n_step50_samples: 0
n_chirp_samples: 3